

Mohammad Raif

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Profile

Final-year B.Tech Computer Science student at Jamia Hamdard with strong interest in **Artificial Intelligence, Data Science, Information Retrieval, and Responsible AI**. Experienced in building applied AI systems involving semantic document analysis, embedding-based retrieval, search pipelines, computer vision, data processing, and full-stack integration. Worked as an **AI Intern** on real-world industry projects including a **Legal PDF Analyzer** and an **Indian Navy object detection project**. Research interests include **bias in recommendation systems, ethical and trustworthy AI**, and intelligent systems for real-world deployment.

Technical Skills

- **Programming Languages:** Python, SQL, JavaScript, C++, HTML5, CSS3
- **AI / Data Science:** Pandas, NumPy, Scikit-learn, Data Preprocessing, Feature Engineering, Model Evaluation
- **Technologies:** FastAPI, REST APIs, Elasticsearch, Embedding-based Search, Jupyter Notebook
- **Databases:** MySQL, MongoDB
- **Tools:** Git, GitHub, VS Code, Excel
- **Core Areas:** Machine Learning, Data Analysis, Information Retrieval, Recommendation Systems, Computer Vision, Data Structures & Algorithms

Education

B.Tech in Computer Science Engineering
Jamia Hamdard, New Delhi

Expected 2026

Experience

AI Intern — Appolo Computers Pvt. Ltd., New Delhi

Dec 2025 – Mar 2026

- Worked on two applied AI projects: a **Legal PDF Analyzer** and an **YOLO-based Object Detection System**, contributing across system architecture, model evaluation, data pipelines, retrieval, and integration.
- Designed and implemented parts of the end-to-end system architecture and data flow for production-oriented AI applications, with emphasis on efficient processing, structured storage, and usability.
- Evaluated multiple machine learning models to identify solutions with improved performance and reduced processing time under deployment-oriented constraints.
- Contributed across backend, data, and interface integration to ensure seamless interaction between model outputs, retrieval components, and user-facing visual layers.

Full Stack Developer Intern — Cosedge

Aug 2025 – Oct 2025

- Worked as a Full Stack Developer intern and developed responsive UI components with focus on frontend usability, interaction flow, and integration with backend APIs.

Full Stack Intern — Eduexcel Global

Feb 2025 – Jul 2025

- Built a responsive platform for international students to explore courses and connect through integrated support and communication features.
- Contributed as a detail-oriented full stack intern, solving development tasks collaboratively and delivering quality work within deadlines.

Research Work

Bias in Recommendation Systems on X and YouTube

- Conducted research on how recommendation models on platforms such as **X** and **YouTube** can create and amplify bias through ranking logic, engagement optimization, and iterative feedback loops.
- Studied **popularity bias**, **exposure bias**, and preference reinforcement, where already dominant content receives disproportionate visibility compared to diverse or emerging content.
- Examined how signals such as clicks, watch time, engagement, and interaction history affect content discoverability, creator reach, and information diversity on large-scale digital platforms.
- Explored the broader societal and technical implications of biased recommendation pipelines, including echo chambers, uneven visibility allocation, and fairness concerns in platform ecosystems.
- Developed a research interest in fairness-aware recommendation design, transparency, accountability, and balanced optimization objectives for responsible recommender systems.

Research in Ethical, Safe, and Responsible AI

- Worked on research related to ethical, safe, and responsible AI with emphasis on AI lifecycle governance and real-world deployment frameworks.
- Presented work titled “**Ethical, Safe & Responsible AI: AI Lifecycle Governance Framework for Real-World Deployment**” at the National Seminar on **AI Unleashed: Innovations, Ethics, and Impact Across Emerging Real-World Domains**.
- Developed academic interest in trustworthy AI, governance frameworks, deployment ethics, and the responsible use of machine learning systems in practice.

Projects

Legal PDF Analyzer

- Developed an AI-powered legal document analysis system for processing uploaded legal PDFs and transforming them into structured, searchable, and interconnected knowledge representations.
- Built and improved **data ingestion pipelines** for extracting and organizing legal document content for downstream retrieval and analysis.
- Implemented **Elasticsearch mappings** for efficient indexing, storage, and retrieval of legal text and metadata.
- Worked on **embedding-based semantic search** and **NLP-powered search functionality** using open-source models to improve contextual retrieval beyond exact keyword matching.
- Integrated **case summarization models** to generate concise summaries of legal cases, enabling faster interpretation of lengthy judicial documents.
- Developed functionality to extract and present important legal outcomes, including identifying **the winning party in a case** along with the generated summary.
- Designed the representation of legal case relationships by showing **parent-child relationships between cases** in a **tree structure**, enabling hierarchical exploration of legal references.
- Supported a **graph-based view** across uploaded documents that visualized how cases were related to one another, functioning like a **knowledge graph** for legal case relationships.
- Contributed to frontend-backend integration and graph/tree visualizations for more intuitive legal document exploration, retrieval, and relationship analysis.
- The project was presented at the **India AI Impact Summit 2026**.

Indian Navy Object Detection Project

- Worked on an object detection pipeline in the context of an **Indian Navy project**, using **YOLO-based models** for video-based detection of ships, humans, and aircraft.
- Performed model evaluation and comparative testing to identify accurate and efficient solutions suitable for real-world operational scenarios.
- Contributed to system architecture and testing workflows for deployment readiness and reliable model behavior.
- Implemented **tracking ID generation** for detected objects to support temporal consistency across video frames.
- Performed **image annotation** and supported model fine-tuning to improve detection quality on domain-specific visual data.

Phishing URL Detection System

- Developed a machine learning-based web application for phishing URL detection using Python and FastAPI.

- Extracted and analyzed URL-based features to classify malicious and legitimate websites in real time.
- Built backend inference functionality and exposed the classification workflow through an API-driven architecture.

Phone Recommendation Tool

- Developed a recommendation-oriented web application that analyzes structured smartphone data and suggests suitable devices based on user preferences.
- Built the frontend using HTML, CSS, and JavaScript, and implemented the backend using Python and FastAPI.
- Processed Excel-based product datasets and designed filtering logic to support data-driven recommendations.

Spotify Clone

- Built a responsive music player interface using HTML, CSS, and JavaScript with modular frontend structure and dynamic media playback.

Leadership & Involvement

IEEE Website Team Member

- Built and maintained the IEEE JHSB website using Bootstrap, HTML, and JavaScript, collaborating through Git-based version control workflows.

Placement Cell Data Team

- Analyzed structured data using Excel to improve workflow organization, reporting accuracy, and operational efficiency.